

now-famous phrase, “Houston, we’ve had a problem here.” The full series of events that followed was nothing short of a Hollywood script – and in fact, was made into a movie starring Tom Hanks – in which the astronauts had to make an emergency move to the Lunar Module and use it to reach Earth safely on April 17, 1970, braving dehydration, lack of oxygen and other extreme circumstances.

NASA went into overdrive fixing the issues and on January 31, 1971, Apollo 14 took off carrying Alan Shepard, Edgar Mitchell and Stuart Roosa. After landing on the moon on February 5, Shepard and Mitchell conducted two EVAs, collecting 42 kgs of rocks and soil for return to Earth, which would be distributed among 16 countries for tests and analysis. Shepard, the first American to perform an EVA in space, also set a new distance-travelled record on the lunar surface of approximately 9,000 feet. On February 9, the Apollo 14 crew landed safely back on Earth without any incident.

Launched on July 26, 1971, Apollo 15 was the first of the Apollo “J” missions capable of a longer stay time on the moon and greater surface mobility. Apart from Alfred Worden, the crew included David Scott and James Irwin who would become the 7th and 8th person to walk on the moon on July 31.

Over three days, the duo explored the moon more than any of their predecessors. Exploration and geological investigations were enhanced by the addition of the Lunar Roving Vehicle (more details about which will be in the next sub-chapter). Scott and Irwin completed a record 18 hours, 37 minutes of exploration, travelled 17.5 miles in the first car that humans have ever driven on the moon, collected more than 170 pounds of lunar samples, set up the ALSEP array, obtained a core sample from about 10 feet beneath the lunar surface, and provided extensive oral descriptions and photographic documentation of geologic features in the vicinity of the landing site.

Before returning to Earth, however, there was another mission: the launching of a Particles and Fields subsatellite into lunar orbit by the Command and Service Module. The subsatellite was designed to investigate the moon’s mass and gravitational variations, particle composition of space near the moon and the interaction of the moon’s magnetic field with that of Earth. After completing this subsatellite mission successfully, Apollo 15 returned to Earth on August 7, 1971.

John Young, Charles Duke and Thomas Mattingly formed the crew of Apollo 16, which blasted off from Earth on April 16, 1972. The Lunar Module carrying Young and Duke touched down on the moon on April 21. In